MILITARY SPECIFICATION

MICONAZOLE NITRATE CREAM, VAGINAL

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Miconazole Nitrate Cream, Vaginal (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Miconazole Nitrate Cream, 2%.

3.1.1 Shall be suitable for the treatment of vaginal moniliasis.

3.1.2 pH. The pH of the finished cream at the time of manufacture shall be not less than 2.6 and not more than 3.0 when determined potentiometrically, undiluted, at 25° C, using the U.S.P. method. During the expiration dating period the pH shall be not less than 2.5 and not more than 3.5 when similarly determined.

3.1.3 Assay. The finished cream shall contain not less than 1.8 percent and not more than 2.2 percent (w/w) of the required amount of miconazole nitrate when determined as specified in 4.8.1.

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3.1.4 Identity. The infrared absorption spectrum of a potassium bromide dispersion of the extracted Miconazole Nitrate shall exhibit maxima only at the same wavelengths as that of a similar dispersion of Miconazole Nitrate Reference Standard.

3.2 Active ingredients.

3.2.1 Miconazole Nitrate.

3.2.1.1 Melting range. Shall melt between 175°C and 185°C, with the beginning and end of melting range not exceeding 3°C, when determined by the U.S.P. Class IA method.

3.2.1.2 Loss on drying. Shall not exceed 0.5 percent when dried at 105°C for two hours.

3.2.1.3 Ultraviolet absorption. The ultraviolet absorption spectrum of a solution containing approximately 40 mg in 100 ml of methanol shall exhibit maxima at 265, 272, and 280 nm ± 2 nm, and the respective absorptivities shall not differ by more than 3.0 percent compared to Miconazole Nitrate Reference Standard concomitantly determined.

3.2.1.4 Infrared spectrum. The infrared absorption spectrum of a potassium bromide dispersion of it previously dried at 105° exhibits maxima only at the same wavelengths as that of a similar preparation of Miconazole Nitrate Reference Standard.

3.2.1.5 Residue on ignition. Shall not exceed 0.2 percent when determined by the U.S.P. method.

3.2.1.6 Foreign related substances. The thin layer chromatograph shall show only one spot, with the same Rf value as that of The Reference Standard, when determined as specified in 4.10.1.

3.2.1.7 Assay. Shall contain not less than 98.0 percent and not more than 102.0 percent of Miconazole Nitrate when determined as specified in 4.10.2.

3.3 Other ingredients. All other ingredients entering into the manufacture of the cream shall be of U.S.P. or N.F. quality, or, if not included in either of these compendia they shall be suitable for use in the cream.

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3.4 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or impair their serviceability.

3.5 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.6 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform and homogeneous quantity of Miconazole Nitrate cream produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 and an acceptable quality level (AQL) of 1.0 percent defective for major defects.

4.4 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the number of units of issues of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.6 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

Table I. For raw material and end item testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
pH	3.1.2	S-2	5 } Composite
Assay	3.1.3	S-2	5 } Composite
Identity	3.1.4	S-2	5 } Composite
Ingredient: Miconazole Nitrate	3.2.1	2/	

1/ Where possible the sample shall be used for two or more tests.

2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.6.2 Samples. If (i) the quality assurance representative submits samples to the laboratory at the Defense Personnel Support Center, or (ii) the bidder (offeror) is required, by the terms of the Schedule or otherwise, to submit samples to said laboratory, then a quantity of each lot of miconazole nitrate used in the manufacture of each lot of cream, sufficient to determine all the required tests in duplicate, and 0.5 gram of miconazole nitrate standard accompanied by a certificate of analysis listing all test results, shall be submitted with the finished product to the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, Attention: Quality Assurance Branch, 2800 South 20th Street, Philadelphia, PA 19101.

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4.7 Examination. The finished cream shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3, 4.4, 4.5, and 4.6.

4.7.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE II. Classification of defects

	Categories and defects 1/
Major	
101	Cream not free from foreign matter.
102	Cream not uniform in color.
103	Cream not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.8 Tests of finished cream.

4.8.1 Assay.

4.8.1.1 Ultraviolet assay.

Standard solution - Accurately weigh approximately 100 mg of Miconazole Nitrate Reference Standard into a 200 ml volumetric flask. Dissolve in, and dilute to volume with, methanol.

Procedure - Transfer approximately 5 grams of Cream accurately weighed into a 100 ml volumetric flask containing 80 ml of isopropyl/chloroform solution (1:1). Mechanically shake to effect solution, and dilute to volume with the preceeding solvent. Pipet 25.0 ml into a beaker and evaporate to dryness on a steam bath with the aid of filtered air or nitrogen. Pipet 10.0 ml of chloroform to the residue and heat just to boiling on the steam bath. Remove the beaker and stir to effect solution (avoid excessive evaporating the chloroform). Add 50 ml of pentane in small portions stirring continuously. Permit to stand 10-15 minutes until the precipitate crystallizes.

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Filter the solution through a medium porosity sintered glass funnel with the aid of filtered air or nitrogen. Wash the beaker with 4 x 5 ml portions of pentane and add the washings to the filter funnel. Similarly rinse the funnel with pentane. Rinse the stem of the funnel with pentane to remove traces of cream base. Dry the precipitate with filtered air or nitrogen.

Reconstitute the precipitate by washing the beaker and funnel repeatedly with small portions of methanol and quantitatively collecting the filtrate in a 50-ml volumetric flask. Dilute to volume with methanol. 1/ Determine the absorbances of standard and sample at the maximum absorption at approximately 271 nm in a 1-cm cell, using methanol as the blank, with a suitable spectrophotometer.

Calculation:

$$\frac{Au}{As} \times \left(\frac{\text{mg standard}}{200}\right) \left(\frac{100}{\text{Weight sample (mg)}} \times \frac{50}{25}\right) \times \frac{100}{1} = \text{percent Miconazole Nitrate (w/w)}$$

where:

Au = absorbance sample
As = absorbance standard

4.8.1.2 Gas chromatographic assay. As an alternate, an appropriate, accurate, reproducible gas chromotographic method shall be used using the final methanol solution obtained in the preceeding assay. 1/

4.9 Identity. Transfer approximately 25 to 30 ml of the final methanol solution prepared in the UV assay 1/ to a beaker and evaporate to dryness on a steam bath with the aid of filtered air or nitrogen. Dry the residue at 105°C for 10 minutes. Prepare a potassium bromide dispersion of approximately 3 mg of the residue in 100 mg of potassium bromide. Determine the spectrum of the pellet from 2 to 16 microns with a spectrophotometer.

4.10 Miconazole Nitrate powder.

4.10.1 Foreign related substances. Prepare a glass silica gel plate by dipping into a solution of 5 grams of NaOH in 250 ml of methanol, and dry.

1/ This solution is retained for the identity test and Gas Chromatographic assay.

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Procedure. Dissolve 30 mg in a mixture of 2 ml of chloroform and one ml methanol. Spot 10 ul of this solution on the plate prepared as above. Spot the same amount of miconazole nitrate reference standard similarly prepared, on the plate adjacent to the sample. Develop the plate using a mixture of 6 volumes of hexane, 3 volumes of chloroform, and one volume of methanol. After drying the plate, place in a chamber containing iodine vapors to visualize the spots.

4.10.2 Assay. Shall be determined by an appropriate, accurate, and reproducible gas chromatographic method. The bidder or offeror shall submit with its bid or proposal the detailed proposed method of assay. Assay methods shall be validated by submission of an adequate number of test results including date of assay, lot number, percentage of required amount, and statistical data including as a minimum the Standard Deviation and Relative Standard Deviation. If the method is approved and accepted by the Government, the method shall be employed in the testing of the active powder, and need not be resubmitted, provided the bidder or offeror certifies that there has been no change in supplier, or specifications for the material.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One package containing one tube along with an applicator, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available tube considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

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5.2.2.3 Intermediate package. Six unit packages shall be packaged in an intermediate package as specified in 5.2.12.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four intermediate packages (144 units) shall be packed in an exterior container as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

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5.4.1.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplier, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II Shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-01-009-6539	MICONAZOLE NITRATE CREAM, VAGINAL, 2%, 85 Gram

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

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6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1507

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-37363(DSA-DM) MICONAZOLE NITRATE CREAM, VAGINAL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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